Economia das Organizações e dos Incentivos — MPE Insper

Exercise List: Real Book Exercises, Lectures 1–4

How to use this list. Every exercise below is a real exercise from one of the course books, reproduced verbatim: the source line gives author, book, chapter, exercise, the printed page and, in brackets, the page your PDF viewer shows for the file in `Livros/`, and whether the source gives the answer. Muñoz-Garcia & Toro-Gonzalez print a worked answer right after each exercise; MWG answers are in the MWG solutions manual in `Livros/`; Tadelis and Gibbons give no answer in the book. When an exercise refers to a model, game or figure stated elsewhere in its book, that material is copied into the same block under "Referenced material", so every exercise can be solved where it is printed. There is one block per lecture, six exercises each, in rising difficulty: two foundations, three exam level, one advanced. Try each exercise in the space provided before opening the book's answer.

Block	#	Source	p. [PDF]	Level	What it trains	Answer
L1	1	Tadelis 4.5	77 [96]	Found.	IESDS round by round, 3×3 matrix	none
	2	Gibbons 1.2	48 [56]	Found.	IESDS, then pure Nash equilibria	none
	3	Tadelis 4.6	77 [96]	Exam	IESDS with continuous strategies	none
	4	Tadelis 4.7	77–78 [96–97]	Exam	Story to matrix, then elimination	none
	5	Muñoz-Garcia Ch. 1 Ex. 12	23 [36]	Exam	Rationality vs. common knowledge of it	book
	6	MWG 8.B.5	262 [142]	Adv.	Cournot by IESDS: two firms vs. three	manual
L2	7	Tadelis 5.6	96 [115]	Found.	Hawk-dove: matrix, pure equilibria	none
	8	Muñoz-Garcia Ch. 3 Ex. 1	86–87 [99–100]	Found.	Chicken: pure and mixed equilibria	book
	9	Tadelis 5.8	96–97 [115–116]	Exam	Cournot with n firms; competitive limit	none
	10	Tadelis 6.8	125 [144]	Exam	Three-player entry; symmetric mixing	none
	11	Gibbons 1.6	49 [57]	Exam	Cournot, asymmetric costs; corner case	none
	12	MWG 8.D.6	264 [143]	Adv.	A mixed equilibrium exists in every 2×2 game	manual
L3	13	Tadelis 8.8	171–172 [190–191]	Found.	Entry deterrence: a non-credible threat	none
	14	Muñoz-Garcia Ch. 4 Ex. 3	160 [172]	Found.	Sequential prisoner's dilemma	book
	15	Tadelis 8.10	172 [191]	Exam	Subgames with a simultaneous stage	none
	16	Tadelis 8.13	173–174 [192–193]	Exam	Hold-up; a posted price as commitment	none
	17	Muñoz-Garcia Ch. 4 Ex. 12	185–186 [197–198]	Exam	Stackelberg with two and three firms	book
	18	Gibbons 2.1	130 [137]	Adv.	Rotten Kid theorem (continuous actions)	none
L4	19	Tadelis 9.1	186–187 [205–206]	Found.	Stage game played twice; threshold δ	none
	20	Gibbons 2.6	133 [140]	Found.	A leader, two simultaneous followers	none
	21	Tadelis 9.4	188 [207]	Exam	Cournot, then access game: sustaining collusion	none
	22	Tadelis 11.2	236 [255]	Exam	Hold-up: investment before an ultimatum	none
	23	Tadelis 11.5	236–237 [255–256]	Exam	Three periods with $\delta_1 \neq \delta_2$	none
	24	Muñoz-Garcia Ch. 6 Ex. 8	303 [314]	Adv.	Three-player Rubinstein, stationary strategies	book

Pages: printed page, then the PDF viewer page in brackets (offsets in `Livros/book-registry.json`; MWG is a two-pages-per-sheet scan). "Answer": *book* = worked answer printed after the exercise; *manual* = MWG solutions manual in `Livros/`; *none* = no answer in the book. Books: Tadelis (2013), *Game Theory: An Introduction*; Gibbons (1992), *Game Theory for Applied Economists*; Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed.; Mas-Colell, Whinston & Green (1995), *Microeconomic Theory* (MWG).

Block 1 — Static games I: dominance and iterated elimination *(Lecture 1)*

Exercise 1. *Iterated Elimination* [Foundations]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.5 "Iterated Elimination", p. 77 [PDF 96]. No answer in the book. Reproduced verbatim.

4.5 Iterated Elimination: In the following normal-form game, which strategy profiles survive iterated elimination of strictly dominated strategies?

		Player 2		
		L	C	R
Player 1	U	6, 8	2, 6	8, 2
	M	8, 2	4, 4	9, 5
	D	8, 10	4, 6	6, 7

Resolution:

Exercise 2. *IESDS and pure-strategy Nash equilibria* [Foundations]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 1 "Static Games of Complete Information", §1.5 Problems (Section 1.1), Problem 1.2, p. 48 [PDF 56]. No answer in the book. Reproduced verbatim; the payoff pair in cell (T, L) is printed "2.0" (comma typo) and is typeset "2, 0".

1.2. In the following normal-form game, what strategies survive iterated elimination of strictly dominated strategies? What are the pure-strategy Nash equilibria?

	L	C	R
T	2, 0	1, 1	4, 2
M	3, 4	1, 2	2, 3
B	1, 3	0, 2	3, 0

Resolution:

Exercise 3. *Roommates* [Exam level]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.6 "Roommates", p. 77 [PDF 96]. No answer in the book. Reproduced verbatim.

4.6 Roommates: Two roommates each need to choose to clean their apartment, and each can choose an amount of time $t_i \geq 0$ to clean. If their choices are t_i and t_j, then player i's payoff is given by $(10 - t_j)t_i - t_i^2$. (This payoff function implies that the more one roommate cleans, the less valuable is cleaning for the other roommate.)

a. What is the best response correspondence of each player i?

b. Which choices survive one round of IESDS?

c. Which choices survive IESDS?

Resolution:

Exercise 4. *Campaigning* [Exam level]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.7 "Campaigning", pp. 77–78 [PDF 96–97]. No answer in the book. Reproduced verbatim.

4.7 Campaigning: Two candidates, 1 and 2, are running for office. Each has one of three choices in running his campaign: focus on the positive aspects of one's own platform (call this a positive campaign $[P]$), focus on the positive aspects of one's own platform while attacking one's opponent's campaign (call this a balanced campaign $[B]$), or finally focus only on attacking one's opponent (call this a negative campaign $[N]$). All a candidate cares about is the probability of winning, so assume that if a candidate expects to win with probability $\pi \in [0, 1]$, then his payoff is π. The probability that a candidate wins depends on his choice of campaign and his opponent's choice. The probabilities of winning are given as follows:

If both choose the same campaign then each wins with probability 0.5.

If candidate i uses a positive campaign while $j \neq i$ uses a balanced one then i loses for sure.

If candidate i uses a positive campaign while $j \neq i$ uses a negative one then i wins with probability 0.3.

If candidate i uses a negative campaign while $j \neq i$ uses a balanced one then i wins with probability 0.6.

a. Model this story as a normal-form game. (It suffices to be specific about the payoff function of one player and to explain how the other player's payoff function is different and why.)

b. Write down the game in matrix form.

c. What happens at each stage of elimination of strictly dominated strategies? Will this procedure lead to a clear prediction?

Resolution:

Exercise 5. *IDSDS and Rationality* [Exam level]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 1 "Dominance Solvable Games", Exercise 12 "IDSDS and Rationality"[A], p. 23 [PDF 36]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only).

Consider the following 3×3 matrix (Fig. 1.30):

		Player 2		
		F	C	B
Player 1	F	0, 5	2, 4	2, 3
	C	2, 3	0, 5	3, 2
	B	5, 0	2, 4	2, 3

Fig. 1.30 Normal-form representation of a three-players game

a. Does any player have a strictly dominant strategy?

b. Does any player have a strictly dominated strategy?

c. Use your results from part (b) to delete players' strictly dominated strategies assuming rationality, but not assuming common knowledge of rationality. [Hint: This means that every player can delete the strategies he finds to be strictly dominated but cannot put himself in the shoes of his opponent to delete the strategies his opponent finds to be strictly dominated.]

d. Use your results from part (b) to delete players' strictly dominated strategies assuming common knowledge of rationality. [Hint: This means that players can put themselves in the shoes of their opponents to delete strategies each other finds to be strictly dominated, as in the standard approach of IDSDS.]

Resolution:

Exercise 6. *Cournot by successive elimination* [Advanced]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 "Simultaneous-Move Games", Exercises, Exercise 8.B.5^{C}, p. 262 [PDF 142, left half]. Answer in the MWG solutions manual (`Livros/`; 8.B.5 is there). Reproduced verbatim. The pointer to Chapter 12 is context; the statement defines the whole model.

8.B.5^{C} Consider the Cournot duopoly model (discussed extensively in Chapter 12) in which two firms, 1 and 2, simultaneously choose the quantities they will sell on the market, q_1 and q_2. The price each receives for each unit given these quantities is $P(q_1, q_2) = a - b(q_1 + q_2)$. Their costs are c per unit sold.

(a) Argue that successive elimination of strictly dominated strategies yields a unique prediction in this game.

(b) Would this be true if there were three firms instead of two?

Resolution:

Block 2 — Static games II: Nash equilibrium and mixed strategies *(Lecture 2)*

Exercise 7. *Hawk-Dove* [Foundations]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 5 "Pinning Down Beliefs: Nash Equilibrium", Exercise 5.6 "Hawk-Dove", p. 96 [PDF 115]. No answer in the book. Reproduced verbatim, with the book's footnote 12.

5.6 Hawk-Dove: The following game has been widely used in evolutionary biology to understand how fighting and display strategies by animals could coexist in a population. For a typical Hawk-Dove game there are resources to be gained (e.g., food, mates, territories), denoted as v. Each of two players can choose to be aggressive, as Hawk (H), or compromising, as Dove (D). If both players choose H then they split the resources but lose some payoff from injuries, denoted as k. Assume that $k > \frac{v}{2}$. If both choose D then they split the resources but engage in some display of power that carries a display cost d, with $d < \frac{v}{2}$. Finally, if player i chooses H while j chooses D then i gets all the resources while j leaves with no benefits and no costs.

a. Describe this game in a matrix.

b. Assume that $v = 10$, $k = 6$, and $d = 4$. What outcomes can be supported as pure-strategy Nash equilibria?[12]

[12] In the evolutionary biology literature, the analysis performed is of a very different nature. Instead of considering the Nash equilibrium analysis of a static game, the analysis is a dynamic one in which successful strategies "replicate" in a large population. This analysis is part of a methodology called evolutionary game theory. For more on the subject see Gintis (2000).

Resolution:

Exercise 8. *Game of Chicken* [Foundations]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 3 "Mixed Strategies, Strictly Competitive Games, and Correlated Equilibria", Exercise 1 "Game of Chicken"[A], pp. 86–87 [PDF 99–100]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only).

Consider the game of Chicken (depicted in Fig. 3.1), in which two players driving their cars against each other must decide whether or not to swerve.

Part (a) Is there any strictly dominated strategy for Player 1? And for Player 2?

Part (b) What are the best responses for Player 1? And for Player 2?

Part (c) Can you find any pure strategy Nash Equilibrium (psNE) in this game?

Part (d) Find the mixed strategy Nash Equilibrium (msNE) of the game. *Hint*: denote by p the probability that Player 1 chooses Straight and by $(1-p)$ the probability that he chooses to Swerve. Similarly, let q denote the probability that Player 2 chooses Straight and $(1-q)$ the probability that she chooses to Swerve.

Part (e) Show your result of part (d) by graphically representing every player i's best response function $BRF_i(s_j)$, where $s_j = \{Swerve, Straight\}$ is the strategy selected by player $j \neq i$.

		Player 2	
		Straight	Swerve
Player 1	Straight	0,0	3,1
	Swerve	1,3	2,2

Fig. 3.1 Game of Chicken

Resolution:

Exercise 9. *The n-Firm Cournot Model* [Exam level]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 5 "Pinning Down Beliefs: Nash Equilibrium", Exercise 5.8 "The n-Firm Cournot Model", pp. 96–97 [PDF 115–116]. No answer in the book. Reproduced verbatim (the printed "$Q = q_i + \cdots + q_n$" is kept as printed).

5.8 The n-Firm Cournot Model: Suppose there are n firms in the Cournot oligopoly model. Let q_i denote the quantity produced by firm i, and let $Q = q_i + \cdots + q_n$ denote the aggregate production. Let $P(Q)$ denote the market clearing price (when demand equals Q) and assume that the inverse demand function is given by $P(Q) = a - Q$, where $Q \leq a$. Assume that firms have no fixed cost and that the cost of producing quantity q_i is cq_i (all firms have the same marginal cost, and assume that $c < a$).

a. Model this as a normal-form game.

b. What is the Nash (Cournot) equilibrium of the game in which firms choose their quantities simultaneously?

c. What happens to the equilibrium price as n approaches infinity? Is this familiar?

Resolution:

Exercise 10. *Market Entry* [Exam level]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 6 "Mixed Strategies", Exercise 6.8 "Market Entry", p. 125 [PDF 144]. No answer in the book. Reproduced verbatim.

6.8 Market Entry: Three firms are considering entering a new market. The payoff for each firm that enters is $\frac{150}{n}$, where n is the number of firms that enter. The cost of entering is 62.

a. Find all the pure-strategy Nash equilibria.

b. Find the symmetric mixed-strategy equilibrium in which all three players enter with the same probability.

Resolution:

Exercise 11. *Cournot with asymmetric costs* [Exam level]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 1 "Static Games of Complete Information", §1.5 Problems (Section 1.2), Problem 1.6, p. 49 [PDF 57]. No answer in the book. Reproduced verbatim.

1.6. Consider the Cournot duopoly model where inverse demand is $P(Q) = a - Q$ but firms have asymmetric marginal costs: c_1 for firm 1 and c_2 for firm 2. What is the Nash equilibrium if $0 < c_i < a/2$ for each firm? What if $c_1 < c_2 < a$ but $2c_2 > a + c_1$?

Referenced material. Source. "the Cournot duopoly model": Gibbons (1992), §1.2.A "Cournot Model of Duopoly", p. 15 [PDF 23], the paragraph that sets up the model. Reproduced verbatim; footnote 6 (a pointer to later sections) omitted.

Let q_1 and q_2 denote the quantities (of a homogeneous product) produced by firms 1 and 2, respectively. Let $P(Q) = a - Q$ be the market-clearing price when the aggregate quantity on the market is $Q = q_1 + q_2$. (More precisely, $P(Q) = a - Q$ for $Q < a$, and $P(Q) = 0$ for $Q \geq a$.) Assume that the total cost to firm i of producing quantity q_i is $C_i(q_i) = cq_i$. That is, there are no fixed costs and the marginal cost is constant at c, where we assume $c < a$. Following Cournot, suppose that the firms choose their quantities simultaneously.

Resolution:

Exercise 12. *Existence in* 2×2 *games* [Advanced]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 "Simultaneous-Move Games", Exercises, Exercise 8.D.6^{B}, p. 264 [PDF 143, left half]. Answer in the MWG solutions manual (`Livros/`; 8.D.6 is there). Reproduced verbatim.

8.D.6^{B} Consider any two-player game of the following form (where letters indicate arbitrary payoffs):

		Player 2	
		b_1	b_2
Player 1	a_1	u, v	ℓ, m
	a_2	w, x	y, z

Show that a mixed strategy Nash equilibrium always exists in this game. [*Hint:* Define player 1's strategy to be his probability of choosing action a_1 and player 2's to be his probability of choosing b_1; then examine the best-response correspondences of the two players.]

Resolution:

Block 3 — Dynamic games and subgame perfection *(Lecture 3)*

Exercise 13. *Entry Deterrence 1* [Foundations]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.8 "Entry Deterrence 1", pp. 171–172 [PDF 190–191]. No answer in the book. Reproduced verbatim.

8.8 Entry Deterrence 1: NSG Corporation is considering entry into the local phone market in the San Francisco Bay Area. The incumbent, S&P Corporation, predicts that a price war will result if NSG enters. If NSG stays out, S&P earns monopoly profits valued at \$10 million (net present value of profits) while NSG earns zero. If NSG enters, it must incur irreversible entry costs of \$2 million. If there is a price war, each firm earns \$1 million. S&P always has the option of accommodating entry (i.e., not starting a price war). In such a case both firms earn \$4 million (net present value). Suppose that the timing is such that NSG first has to choose whether or not to enter the market. Then S&P decides whether to "accommodate entry" or "engage in a price war." What is the subgame-perfect equilibrium outcome to this sequential game? Set up a game tree.

Resolution:

Exercise 14. *Sequential Version of the Prisoner's Dilemma Game* [Foundations]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 4 "Sequential-Move Games with Complete Information", Exercise 3 "Sequential Version of the Prisoner's Dilemma Game"[A], p. 160 [PDF 172]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only); Fig. 4.9 cropped from the page.

Consider the game tree in Fig. 4.9, representing the sequential-move version of the Prisoner's dilemma game, where player 1 acts first, and observing its move player 2 responds.

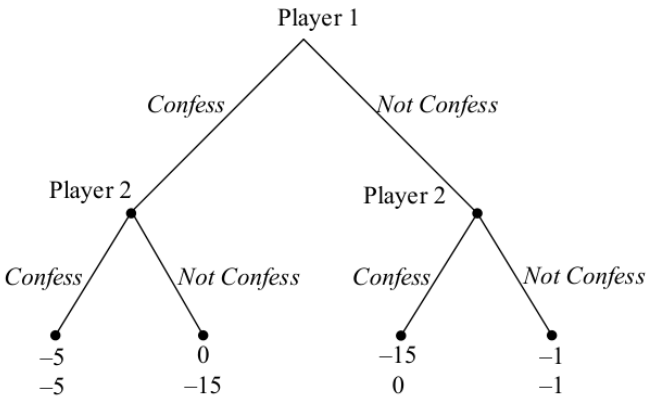


Fig. 4.9 Prisoner's dilemma sequential-move game

(a) Find the best responses of the second mover, for each of the first mover's actions.

(b) Find the first mover's equilibrium action and describe the subgame perfect Nash equilibrium of the game.

(c) Does the equilibrium behavior in the sequential-move game differ from its simultaneous-move version?

Referenced material. Source. "its simultaneous-move version": Muñoz-Garcia & Toro-Gonzalez (2019), Ch. 1, Exercise 5 "Prisoners' Dilemma Game", Fig. 1.8, p. 9 [PDF 22]. Matrix rebuilt from the page.

		Prisoner 2: Confess	Prisoner 2: Not confess
Prisoner 1	Confess	-5, -5	0, -15
	Not confess	-15, 0	-1, -1

Fig. 1.8 Normal-form of Prisoners' dilemma game

Resolution:

Exercise 15. *Playing It Safe* [Exam level]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.10 "Playing It Safe", p. 172 [PDF 191]. No answer in the book. Reproduced verbatim.

8.10 Playing It Safe: Consider the following dynamic game: Player 1 can choose to play it safe (denote this choice by S), in which case both he and player 2 get a payoff of 3 each, or he can risk playing a game with player 2 (denote this choice by R). If he chooses R then they play the following simultaneous-move game:

		Player 2: A	Player 2: B
Player 1	C	8, 0	0, 2
	D	6, 6	2, 2

a. Draw a game tree that represents this game. How many proper subgames does it have?

b. Are there other game trees that would work? Explain briefly.

c. Construct the matrix representation of the normal form of this dynamic game.

d. Find all the Nash and subgame-perfect equilibria of the dynamic game.

Resolution:

Exercise 16. *Junk Mail Advertising* [Exam level]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.13 "Junk Mail Advertising", pp. 173–174 [PDF 192–193]. No answer in the book. Reproduced verbatim.

8.13 Junk Mail Advertising: Suppose a single good is owned by a single seller who values it at $c > 0$ (he can consume the good and get a payoff of c). There is a single buyer who has a small transportation cost $k > 0$ to get to and back from the seller's store, and he values the good at $v > c + k$. The buyer first decides whether to make the commute or stay at home, not buy the good, and receive a payoff of 0. If the buyer commutes to the store, the seller can then make the buyer a take-it-or-leave-it price offer $p \geq 0$. The buyer can then accept the offer, pay p, and get the good, or he can walk out and not buy the good. Assume that c, v, and k are common knowledge.

a. As best you can, draw the extensive form of this game. What is the best response of the buyer at the node at which he decides whether to accept or reject the seller's offer?

b. Find the subgame-perfect equilibrium of the game and show that it is unique. Is it Pareto optimal?

c. Are there Nash equilibria that yield a higher payoff to both players as compared to the subgame-perfect equilibrium you found in (b)?

d. Now assume that before the game is played the seller can, at a small cost $\varepsilon < (v - c - k)$, send the buyer a postcard that commits the seller to a certain price at which the buyer can buy the good (e.g., "bring this coupon and get the good at a price p"). Would the seller choose to do so? Justify your answer with an equilibrium analysis.

Resolution:

Exercise 17. *Stackelberg Competition with 2, or 3 Firms* [Exam level]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 4 "Sequential-Move Games with Complete Information", Exercise 12 "Stackelberg Competition with 2, or 3 Firms"[B], pp. 185–186 [PDF 197–198]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only).

Consider a Stackelberg model of output competition where firm 1 is the industry leader and firm 2 is the follower. Firms face a linear inverse demand curve $p(Q) = 10 - Q$, where $Q = q_1 + q_2$ denotes the aggregate output sold by both firms. In addition, consider that firms are cost symmetric, facing a common marginal cost of $MC_i = 2$ for every firm $i = \{1, 2\}$ and no fixed costs.

(a) Find the equilibrium in this Stackelberg game.

(b) Assume now that a new firm (firm 3) joins the industry, acting last (after firm 2 chooses its output). Find the equilibrium in this Stackelberg game. Compare your results with those in part (a). How is the output of firm 1 and 2 affected by firm 3's entry?

Resolution:

Exercise 18. *The Rotten Kid Theorem* [Advanced]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 "Dynamic Games of Complete Information", §2.6 Problems (Section 2.1), Problem 2.1, p. 130 [PDF 137]. No answer in the book. Reproduced verbatim.

2.1. Suppose a parent and child play the following game, first analyzed by Becker (1974). First, the child takes an action, A, that produces income for the child, $I_C(A)$, and income for the parent, $I_P(A)$. (Think of $I_C(A)$ as the child's income net of any costs of the action A.) Second, the parent observes the incomes I_C and I_P and then chooses a bequest, B, to leave to the child. The child's payoff is $U(I_C + B)$; the parent's is $V(I_P - B) + kU(I_C + B)$, where $k > 0$ reflects the parent's concern for the child's well-being. Assume that: the action is a nonnegative number, $A \geq 0$; the income functions $I_C(A)$ and $I_P(A)$ are strictly concave and are maximized at $A_C > 0$ and $A_P > 0$, respectively; the bequest B can be positive or negative; and the utility functions U and V are increasing and strictly concave. Prove the "Rotten Kid" Theorem: in the backwards-induction outcome, the child chooses the action that maximizes the family's aggregate income, $I_C(A) + I_P(A)$, even though only the parent's payoff exhibits altruism.

Resolution:

Block 4 — Multistage games and bargaining *(Lecture 4)*

Exercise 19. *Multistage Practice* [Foundations]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 9 "Multistage Games", Exercise 9.1 "Multistage Practice", pp. 186–187 [PDF 205–206]. No answer in the book. Reproduced verbatim.

9.1 Multistage Practice: Consider the following simultaneous-move game that is played twice (the players observe the first-period outcome prior to the second-period play):

		Player 2		
		L	C	R
Player 1	T	10, 10	2, 12	0, 13
	M	12, 2	5, 5	0, 0
	B	13, 0	0, 0	1, 1

a. Find all the pure-strategy subgame-perfect equilibria with no discounting ($\delta = 1$). Be precise in defining history-contingent strategies for both players.

b. For each of the equilibria you found in (a), find the smallest discount factor that supports it.

Resolution:

Exercise 20. *One leader, two followers* [Foundations]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 "Dynamic Games of Complete Information", §2.6 Problems (Section 2.2), Problem 2.6, p. 133 [PDF 140]. No answer in the book. Reproduced verbatim.

2.6. Three oligopolists operate in a market with inverse demand given by $P(Q) = a - Q$, where $Q = q_1 + q_2 + q_3$ and q_i is the quantity produced by firm i. Each firm has a constant marginal cost of production, c, and no fixed cost. The firms choose their quantities as follows: (1) firm 1 chooses $q_1 \geq 0$; (2) firms 2 and 3 observe q_1 and then simultaneously choose q_2 and q_3, respectively. What is the subgame-perfect outcome?

Resolution:

Exercise 21. *Online Gaming* [Exam level]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 9 "Multistage Games", Exercise 9.4 "Online Gaming", p. 188 [PDF 207]. No answer in the book. Reproduced verbatim.

9.4 Online Gaming: Consider a two-stage game between two firms that produce online games. In the first stage, they play a Cournot competition game (each chooses a quantity q_i) with demand function $p = 100 - q$ and zero marginal production costs ($c_i(q_i) = 0$ for $i = 1, 2$). In the second stage, after observing the pair (q_1, q_2) and after profits have been distributed, the players play a simultaneous-move "access" game in which each can open its platform to allow players on the other platform to play online with players on its platform (O for player 1, o for player 2), or choose to keep its platform noncompatible (N for player 1, n for player 2), in which case each platform's players can play only with others on the same platform. If they choose (N, n) then second-stage payoffs are $(0, 0)$. If only one firm chooses to open its platform, it bears a cost of -10 with no benefit, since the other firm did not allow open access. Finally if both firms choose (O, o) then each firm gets many more eyeballs for advertising, and payoffs for each are 2500. Both players use the same discount factor δ to discount future payoffs.

a. Find the unique Nash equilibrium in the first-stage Cournot game and all of the pure-strategy Nash equilibria of the second-stage access game. Find all the pure-strategy subgame-perfect equilibria with extreme discounting ($\delta = 0$). Be precise in defining history-contingent strategies for both players.

b. Now let $\delta = 1$. Find a subgame-perfect equilibrium for the two-stage game in which the players choose the monopoly (total profit–maximizing) quantities and split them equally (a symmetric equilibrium).

c. What is the lowest value of δ for which the subgame-perfect equilibrium you found in (b) survives?

d. Now let $\delta = 0.4$. Can you support a subgame-perfect equilibrium for the two-stage game in which the players choose the monopoly quantities and split them equally? If not, what are the highest profits that the firms can make in a symmetric equilibrium?

Resolution:

Exercise 22. *Holdup* [Exam level]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 11 "Strategic Bargaining", Exercise 11.2 "Holdup", p. 236 [PDF 255]. No answer in the book. Reproduced verbatim.

11.2 Holdup: Consider an ultimatum game ($T = 1$ bargaining game) in which before player 1 makes his offer to player 2, player 2 can invest in the size of the pie. If player 2 chooses a low level of investment (L) then the size of the pie is small, equal to v_L, while if player 2 chooses a high level of investment (H) then the size of the pie is large, equal to v_H. The cost to player 2 of choosing L is c_L, while the cost of choosing H is c_H. Assume that $v_H > v_L > 0$, $c_H > c_L > 0$, and $v_H - c_H > v_L - c_L$.

a. What is the unique subgame-perfect equilibrium of this game? Is it Pareto optimal?

b. Can you find a Nash equilibrium of the game that results in an outcome that is better for both players as compared to the unique subgame-perfect equilibrium?

Referenced material. Source. "ultimatum game ($T = 1$ bargaining game)": Tadelis (2013), Ch. 11, the first round of the bargaining game, p. 221 [PDF 240], and §11.1 "One Round of Bargaining: The Ultimatum Game", p. 222 [PDF 241]. Reproduced verbatim (excerpts).

In the first round:

- Player 1 offers shares $(x, 1 - x)$, where player 1 receives x and player 2 receives $1 - x$.
- Player 2 then chooses to accept, causing the game to end with payoffs $v_1 = x$ and $v_2 = 1 - x$, or reject, causing the game to move to the second round.

[. . .] We begin the analysis with an easy case in which there is only one round of bargaining, so that $T = 1$. This is often referred to in the literature as an ultimatum game: player 1 makes an ultimatum take-it-or-leave-it offer to player 2, and player 2 either accepts it, in which case the pie is split between the players, or rejects it, in which case both players receive a payoff of zero.

Resolution:

Exercise 23. *Asymmetric Patience 1* [Exam level]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 11 "Strategic Bargaining", Exercise 11.5 "Asymmetric Patience 1", pp. 236–237 [PDF 255–256]. No answer in the book. Reproduced verbatim.

11.5 Asymmetric Patience 1: Consider a three-period sequential (alternating-offer) bargaining model in which two players have to split a pie worth 1 (starting with player 1 making the offer). Now the players have different discount factors, δ_1 and δ_2.

a. Compute the outcome of the unique subgame-perfect equilibrium.

b. Show that when $\delta_1 = \delta_2$ player 1 has an advantage.

c. What conditions on δ_1 and δ_2 give player 2 an advantage? Why?

Referenced material. Source. the alternating-offer bargaining model this exercise modifies ("Now the players have different discount factors"): Tadelis (2013), Ch. 11 "Strategic Bargaining", p. 221 [PDF 240]. Reproduced verbatim (excerpts); with $\delta_1 \neq \delta_2$, player i discounts with δ_i.

In the first round:

- Player 1 offers shares $(x, 1-x)$, where player 1 receives x and player 2 receives $1-x$.
- Player 2 then chooses to accept, causing the game to end with payoffs $v_1 = x$ and $v_2 = 1-x$, or reject, causing the game to move to the second round.

In the second round:

- A share $1-\delta$ of the pie is dissipated (the pie is discounted with discount factor $0 < \delta < 1$).
- Player 2 offers shares $(x, 1-x)$, where player 1 receives x and player 2 receives $1-x$.
- Player 1 chooses to accept, causing the game to end with payoffs $v_1 = \delta x$ and $v_2 = \delta(1-x)$, or reject, causing the game to move to the third round.

In the third round and beyond:

- The game continues in the same way, where following rejection in an odd period player 2 gets to offer in the next even period, and vice versa.
- Each period has further discounting of the pie, so that in period t the total pie is worth δ^{t-1}.

[...] Assume for now that the game ends in some prespecified final period T, which will be reached if the players did not reach an agreement in the $T-1$ periods that preceded T. Furthermore assume that period T is a "hard deadline," so that both players receive a payoff of zero if they fail to reach an agreement in this last period.

Resolution:

Exercise 24. *Infinite-Horizon Bargaining Game Between Three Players* [Advanced]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 6 "Repeated Games and Correlated Equilibria", Exercise 8 "Infinite-Horizon Bargaining Game Between Three Players"[C], p. 303 [PDF 314]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only).

Consider a three-player variant of Rubinstein's infinite-horizon bargaining game, where players divide a surplus (pie) of size 1, and the set of possible agreements is

$$X = \{(x_1, x_2, x_3) \mid x_i \geq 0 \quad \text{for } i = 1, 2, 3 \ \text{ and } \ x_1 + x_2 + x_3 = 1\}.$$

Players take turns to offer agreements, starting with player 1 in round $t = 0$, and the game ends when *both* responders accept an offer. If agreement x is reached in round t, player i's utility is given by $\delta^t x_i$, where $\frac{1}{2} < \delta < 1$. Show that, if each player is restricted to using stationary strategies (in which she makes the same proposal every time she is the proposer, and uses the same acceptance rule whenever she is a responder), there is a unique subgame perfect equilibrium outcome. How much does each player receive in equilibrium?

Resolution: